



DATA SCIENCE FIELD NOTES / 04A

MACHINE LEARNING PROBLEM DEFINITION

Improving appointment attendance through responsible, pre-appointment support

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PROGRAMME	AnalystLab Africa - Data Science Internship
PROJECT	HealthConnect No-Show Prediction
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Educational fictional-clinic project. Not a clinical decision system or validated operational tool.

Executive summary

HealthConnect is a fictional outpatient clinic seeking to reduce missed appointments and improve patient support. This report defines the machine-learning problem and assesses whether the supplied appointment history can support a responsible no-show prediction prototype.

Provisional decision: use supervised binary classification for an initial attended-versus-no-show experiment, preserve cancellations separately, and use only information available before the appointment.

Measure	Verified value
Appointment records	5,000
Variables	18
Anonymised patient IDs	1,696
No-shows	2,423
Attended	2,314
Cancelled	263
Initial binary cohort	4,737 appointments
No-show rate in binary cohort	51.15%

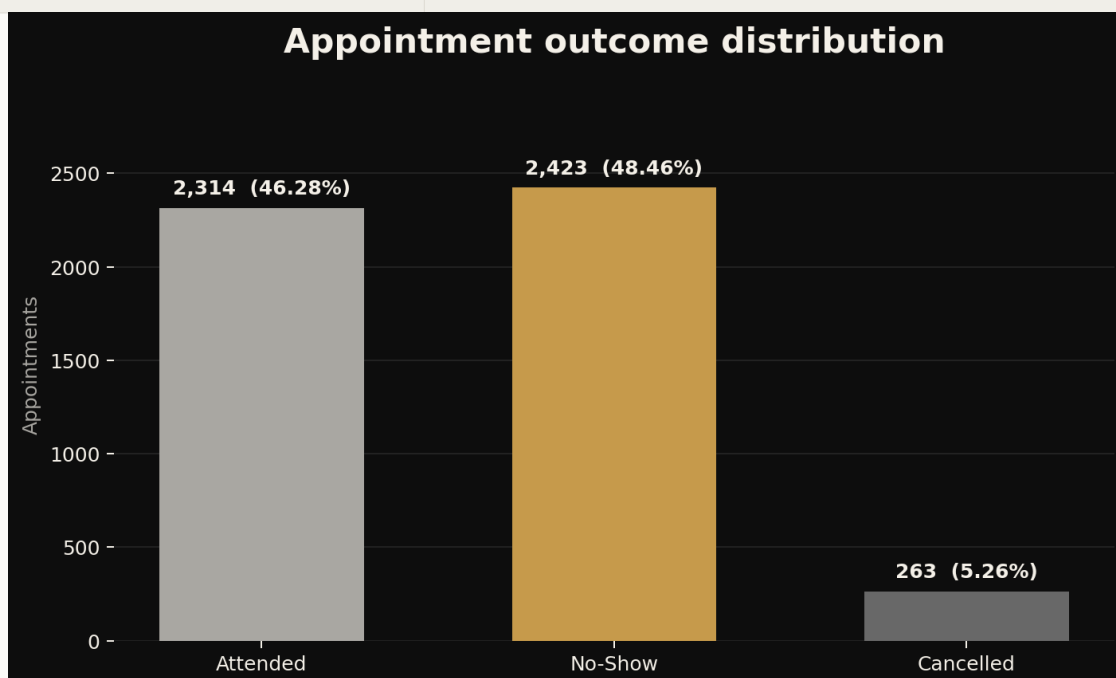


Figure 1. Appointment outcomes in the supplied fictional dataset.

Business understanding

Missed appointments can leave clinical capacity underused, disrupt scheduling and limit organised patient support. The objective is to determine whether information known before an appointment can identify cases that may benefit from confirmation, reminder follow-up or rescheduling assistance.

Stakeholders and intended use

Stakeholder	Interest and responsibility
Patients	Fair, clear and non-punitive attendance support
Administrative staff	Practical use of risk information within available capacity
Healthcare professionals	More predictable schedules and appropriate use of clinical time
Clinic management	Capacity use, governance and operational evaluation
Data and ML teams	Development, validation, monitoring and documentation

The proposed prediction is not an instruction to deny care, deprioritise patients or make clinical decisions. Any later intervention must be supportive, transparent and subject to human review.

Analytical approach

Element	Definition
Learning type	Supervised learning
Primary task	Binary classification: attended versus no-show
Unit of analysis	One scheduled appointment
Positive class	No-Show
Proposed output	No-show probability and an operationally selected threshold
Prediction point	After booking and before the appointment, at a clinic-defined time

Business problem: missed appointments. Data-science problem: prediction. Business action: a separately designed attendance-support workflow informed by the prediction.

Data requirements and collection

- A known appointment outcome and stable appointment identifier.
- A reliable prediction timestamp and feature-availability map.
- Scheduling and booking information available before the prediction point.
- Historical attendance counts calculated strictly before the current appointment.
- Documented categories, units, missing values and date conventions.
- Intervention and outcome data for later operational evaluation.

The CSV and dictionary provide structured fictional records, but reminder and cancellation timestamps, clinic location per appointment, intervention outcomes and an authoritative feature-availability map are absent.

Initial data assessment

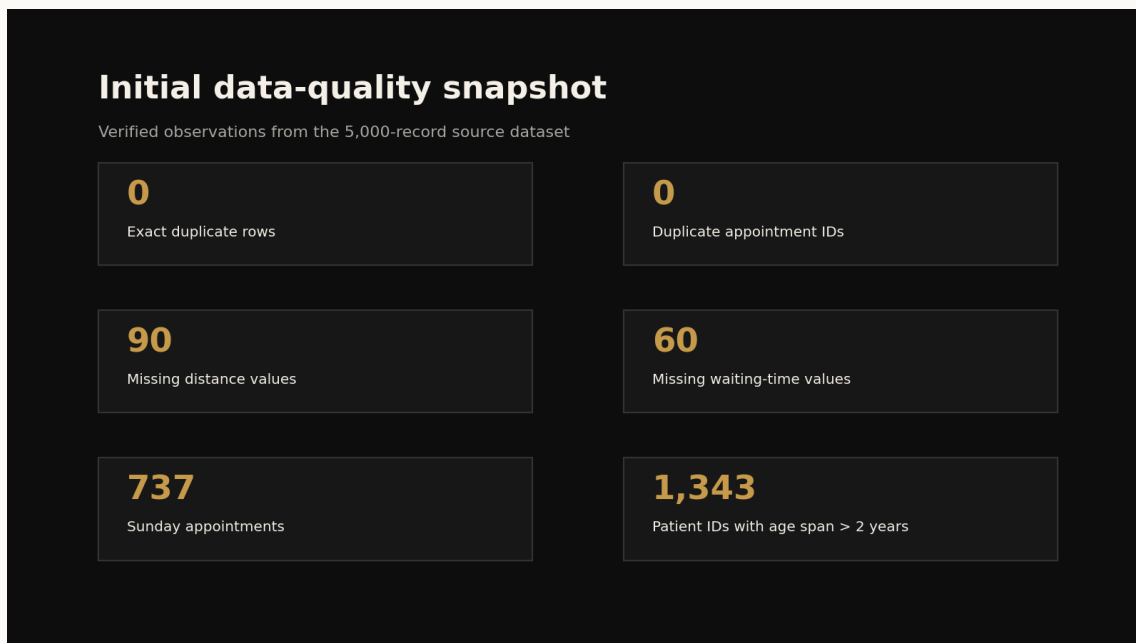


Figure 2. Verified initial data-quality findings.

Issue	Evidence	Current response
Missing distance	90 records	Investigate pattern; fit treatment on training data only
Missing waiting time	60 records	Defer until prediction-time availability is confirmed
Reminder category None	1,366 records	Retain as no reminder channel
Date-format conflict	CSV differs from dictionary	Parse explicitly and document format
Sunday scheduling	737 appointments	Seek clarification; do not silently delete
Repeated-ID age	1,343 IDs span more than two years	Avoid patient-ID prediction

Target and cancellation handling

The initial experiment encodes No-Show as 1 and Attended as 0. The 263 cancelled appointments remain in the raw data but are excluded from the first binary cohort. A cancellation is a different operational event and should not be silently relabelled.

Candidate feature families

Family	Examples	Primary check
Scheduling	Day, month, appointment type	Available before appointment
Booking	Lead days and channel	Correct derivation and timing
History	Previous appointments and no-shows	Calculated before current record
Reminder	Sent flag and channel	Timestamp and causal interpretation
Accessibility	Distance and transport access	Missingness and fairness
Operational	Waiting time	Creation time must be confirmed

Appointment outcome is the target and must never be used as an input. Appointment and patient identifiers are excluded as ordinary predictors. Waiting-time and reminder variables remain conditional on timing and leakage review.

Modelling and evaluation plan

- Define the cohort and prediction time before transforming data.
- Use a future-oriented split when dates support it.
- Fit imputers, encoders and scalers using training data only.
- Compare a simple baseline with an interpretable logistic-regression model.
- Evaluate recall, precision, F1, confusion matrices, ROC-AUC, PR-AUC and calibration.
- Choose thresholds using clinic priorities and staff capacity.
- Review subgroup performance and error differences.

Risks and dependencies

- The data is fictional and cannot establish real-world performance.
- Feature timing is uncertain for reminders, cancellations and waiting time.
- Associations must not be described as causes of patient behaviour.
- Repeated patient histories contain inconsistencies.
- Demographic and accessibility fields require necessity and fairness review.
- The clinic must define the intervention, capacity and feedback loop.

Conclusion

The supplied resources are suitable for an educational prototype, but not operational deployment. The next phase should resolve the quality questions, confirm feature availability, create a leakage-safe cohort and establish a responsible evaluation design.